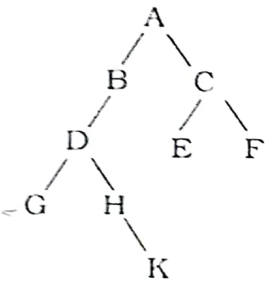
End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech.AI & ML, Semester:2
(July, 2023)

Subject Code: BAM 102	**Subject: Fundamentals of Data Structure**
Time :3 Hours	**Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	
	(a) Differentiate between Row Major and Column Major array Index notation.		
	(b) Explain the complexity of algorithms and time-space tradeoff.		
	(c) Compare a singly linked list with a circular linked list.		
	(d) Differentiate between a Queue and Priority Queue.		
	(e) What are AVL Trees?		
	(f) Discuss the worst and best-case time complexity of Merge Sort with an example.		
	(g) Illustrate with an example the concept of Topological Sort.		
	(h) What are Directed Graphs?		
	UNIT-I		
Q2	(a) What is an array? How it is represented in memory? (b) Write an algorithm/pseudocode to perform the following operations: • Insertion (At the End) in an array • Deletion (At the front) in an array	**(5+5=10)**	
Q3	Define the Data structure. Discuss any two Data Structures in detail.	**(10)**	
	UNIT-II		
Q4	(a) Write an algorithm/pseudocode using an array to implement PUSH and POP operations over the stack. Analyze its time complexity. (b) Convert the following Infix expression to Prefix and Postfix: A + B + D * E / F + B * C	**(5+5=10)**	
Q5	(a) Evaluate the following postfix expression using Stack (Note that ^ is an exponential operator): 8 2 3 ^ / 2 3 * + 5 1 * - (b) Write an algorithm/pseudocode to insert a node at the end of a single linked list.	**(5+5=10)**	
	UNIT-III		
Q6	(a)Write an algorithm/pseudocode to search and insert an element in Binary Search Tree. Analyze its time complexity. (b) Find the Inorder, Postorder, and Preorder Traversal of the given Binary Tree: 	**(5+5=10)**	
Q7	(a) Show the result of inserting 3, 10, 8, 7, 2, 12, 5, 9, 65,18 into an empty Binary Search Tree. Also, show the result of deleting the root. (b) Write an algorithm/ pseudo code for Insertion Sort. Analyze its time complexity.	**(5+5=10)**	
	UNIT-IV		
Q8	Discuss the following with examples: (a) Collision Resolution Techniques (b) Spanning Trees	**(5+5=10)**	
Q9	Give any two representations of graphs. What do you mean by in-degree and out-degree of a graph? Give algorithm/pseudocode for the Dijkstra algorithm. Illustrate the working of the Dijkstra algorithm with a suitable example. Discuss its best-case and worst-case time complexity.	**(1+1+4+2+2=10)**	

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